



# Natural language processing course project report

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## Abstract

We built an AI assistant for the UL FRI student office that answers administrative questions in Slovenian and English. The system is a RAG pipeline that crawls and parses UL/FRI web content, chunks it with a section-aware splitter, embeds the chunks with a multilingual sentence-transformer, and stores them in FAISS. At query time, we compare three retrieval modes (dense, BM25+dense hybrid, and hybrid with a cross-encoder reranker) as inputs to an instruction-tuned generator. We evaluate the pipeline with both classical retrieval metrics ( $\text{Recall}@k$ ,  $\text{MRR}@k$ ,  $\text{nDCG}@k$ ) and a RAGAS-style LLM-as-judge protocol scoring faithfulness, hallucination, refusal handling, and overall quality. Early on we noticed that many of our reference answers were not grounded in the corpus, so we first ran a Sonnet-graded audit and used the cleaned set for every comparison. The automated LLM-judge sweep favors `multilingual-e5-base` with hybrid retrieval, a cross-encoder reranker,  $\text{top-}k=4$ , and 400-token chunks. However, because we treat the final human evaluation as decisive, we select `bge-m3` with hybrid + rerank,  $k=6$ , and reranker pool  $k_r=30$  as the final operating setup (revised after the reference-answer audit, see Human evaluation section). Under this setup, `Llama-3.1-8B-Instruct` is the best generator at 0.665 faithfulness with 7/10 correct refusals on out-of-corpus probes.

## Keywords

Retrieval augmented generation, dense retrieval, hybrid retrieval, multilingual embeddings, student affairs chatbot

## Introduction

We built an assistant for the student office of the Faculty of Computer and Information Science (UL FRI) to answer everyday administrative questions about study programmes, regulations, and office procedures in Slovenian and English. Large language models alone are a poor fit for this task: answers often require precise details (deadlines, paragraph numbers, application forms) that must come from official documents rather than the model's training data. For this reason we used Retrieval Augmented Generation [1], which has proven useful for university-style FAQ chatbots [2, 3] and helps reduce hallucinations and stale knowledge in educational deployments [4]. The pipeline follows the comparative-retrieval setup (dense vs. hybrid vs. reranked, with a chunk-size sweep) used in prior RAG-for-admissions studies [5, 6], and we evaluate locally hosted instruction-tuned LLMs following work on open-day chatbots and RAG vs. fine-tuning for knowledge-heavy tasks [7, 8].

## Data collection and parsing

The pipeline is organised into *runs*, each run contains raw downloads, parsed JSONL, chunks, and a FAISS index in a single directory. A lightweight crawler walks breadth-first from a curated seed list with separate depth limits for `fri.uni-lj.si`, other `*.uni-lj.si` domains, and external domains. It downloads HTML and linked PDF/DOCX, augments results with sitemaps and RSS/Atom feeds, respects `robots.txt`, deduplicates by SHA-256, and reuses `ETag/If-Modified-Since` headers on updates. Each document is stored as a JSON record (URL-derived `doc_id`, title, language, ordered sections): HTML via BeautifulSoup + `trafilatura`, PDF via `Docling`+`Tesseract` or `PyMuPDF`, and DOCX via `python-docx`.

## Chunking, embeddings and retrieval

We split documents with a section-aware sliding window measured in whitespace tokens (default 400, overlap 80). Every chunk is prefixed with the document title and section name, gets a stable `chunk_id`, and is then deduplicated with a SimHash filter. For retrieval experiments we use multilingual sentence-transformer embedders (`intfloat/multilingual-e5-base`,

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intfloat/multilingual-e5-large, and BAAI/bge-m3) and index chunks in FAISS as an IndexFlatIP (cosine under normalisation). At query time we support three retrieval modes: dense (top- $k$  from FAISS), hybrid (BM25 from rank\_bm25 with default  $k_1=1.5$ ,  $b=0.75$ , fused with dense scores after min-max normalisation on the union of both top lists), and hybrid plus cross-encoder rerank using mmarco-mMiniLMv2-L12-H384-v1 on a candidate pool of size 20. Retrieved chunks are then formatted into a numbered context block and a language-specific system prompt tells the generator to answer *only* from that block and in the detected language.

### Evaluation set and reference audit

Our evaluation set is a hand-written JSONL file of 61 questions: 51 *in-scope* (the corpus contains a factual answer) and 10 *negatives* (deliberate out-of-corpus probes used to test refusal), shipped with the repository. Each entry carries the expected language, a list of expected keywords, and a free-form `reference_answer` that we use as the human-written gold for the LLM-as-judge protocol below. A smaller development corpus of 25 questions also carries `relevant_doc_ids` (manually annotated) which we use for the document-level retrieval metrics.

The first time we ran the judge, in-scope faithfulness was very low. Inspection showed many reference answers included details not present in the corpus, which caused the judge to mark them down. To prevent this reference drift from affecting later comparisons, we performed a Sonnet-graded fact-check before the configuration sweeps: for each positive question we retrieved a top-30 BM25  $\cup$  dense pool from the current index and asked Claude Sonnet (claude-sonnet-4-6,  $T=0$ ) to label every claim in the reference as `supported`, `partial`, or `unsupported`, and to optionally produce a grounded rewrite. The application policy was fixed: `supported` retained the original, `partial` or `unsupported` with a rewrite used the rewrite (flagged `auto_rewritten`), `unsupported` without a rewrite was dropped, and negatives were never modified. The cleaned set was written as a versioned artefact `questions_full_v2.json`.

### LLM-as-judge protocol

We used a RAGAS-style judge: for each row we sent the question, the retrieved top- $k$  chunks, the model answer, and the reference answer to Claude with a strict-JSON rubric covering *faithfulness*, *answer relevance*, and *context relevance* (each in  $[0, 1]$ ), a binary *hallucination* flag (set when at least one claim was unsupported by the retrieved context), a *refusal* flag, and an integer *overall* score in  $[1, 5]$ . Negatives were scored under the same rubric, with a correct refusal rewarded with high faithfulness. All judge calls ran at  $T=0$  for reproducibility. To contain costs while maintaining quality at the bottleneck, we used Sonnet (claude-sonnet-4-6) for the reference audit and final re-judgment, and Haiku (claude-haiku-4-5) for

the comparison sweeps. The two models agreed to within 0.01 in in-scope faithfulness on the winning configuration (0.739 vs. 0.748); we therefore used Haiku for ranking. Per-configuration indices were built by driver scripts that passed `EMBEDDING_MODEL/CHUNK_SIZE/CHUNK_OVERLAP` as environment overrides and persisted them next to the index for safety on load.

## Results

### Corpus, evaluation set, and reference audit

At the default 400/80 chunk policy the full crawl yields 2,742 parsed documents and 29,296 chunks, up from the 176 documents in the initial *Test* corpus we used during development. Building the index for the full crawl with Docling layout-based OCR took roughly 4 hours. To speed up development we experimented with a two-pass pipeline (PyMuPDF text extraction for standard PDFs, with Tesseract OCR via Docling as a fallback for near-empty pages), which reduced indexing time to about  $\sim 14$  minutes for the same  $\sim 500$  PDFs. We used the faster two-pass pipeline for most experiments, but for the final tests — and for the FAISS index distributed alongside the README — we used the full Docling OCR output so scanned pages and layout text were preserved. Images embedded in documents remain outside the LLM’s descriptive scope. Running the Sonnet audit on the original references against this corpus showed that only 9/47 ( $\sim 19\%$ ) of the in-scope references were fully supported as written. Of the rest, 31 were classified `partial` and rewritten using only chunk-attested facts, 7 were `unsupported` (3 dropped, 4 rewritten where Sonnet found alternative chunks), and 14 were skipped (no factual reference, or negative). The cleaned `questions_full_v2.json` contains 51 in-scope (of which 44 have a written reference answer) and 10 negatives, and is the basis for everything that follows.

### Document-level retrieval (development corpus)

On the smaller 25-question development corpus, where we have document-level annotations, dense retrieval at  $k=4$  already reaches 88.0% Recall@4 and 66.3% nDCG@4. Adding the cross-encoder reranker lifts these to 100.0% and 93.4%, while hybrid + rerank is essentially tied (100.0% / 93.9%). The corresponding chunk-size sweep on the same corpus showed sizes 200–500 all reaching  $\sim 97.9\%$  Recall, with 300 giving the highest precision. The LLM-judged sweep on the full crawl, reported below, revisits this choice on a stricter metric and ends up picking 400.

### Embedder $\times$ retrieval-mode sweep (LLM-as-judge)

Table 1 shows the embedder  $\times$  retrieval-mode sweep at top- $k=4$  on the cleaned set, judged by Claude Haiku at  $T=0$ ; refusal-side faithfulness was  $\geq 0.90$  across all six configurations. Two findings are noteworthy. First, the cross-encoder reranker was the largest single contributor: on `e5-base` it increased faithfulness from 0.575 (dense) and 0.565 (hybrid) to 0.739, and context relevance from about 0.72 to 0.857.

Second, `e5-base + rerank` outperformed `e5-large + rerank` on every in-scope metric while embedding approximately  $2.5\times$  faster, an unexpected result. Once reranking was applied, the smaller first-stage embedder proved sufficient.

### Chunk size on the winning configuration

Holding the winner fixed (`e5-base`, hybrid + rerank,  $k=4$ ) and varying the chunk size with 20% overlap, the Haiku-judged in-scope faithfulness is 0.691 at 200/40 (50,546 chunks), 0.698 at 300/60 (36,331), and 0.739 at 400/80 (29,296). Larger chunks improved performance because the cross-encoder benefited from additional local context per chunk; 400-token chunks also produced the smallest and fastest index of the three, so we selected them as the default.

### Reference-grounding ceiling and generator comparison

We re-judged the winning retrieval configuration with Claude Sonnet at  $T=0$  in two ways. First, as a *reference-grounding ceiling*: given the retrieved top-4 chunks, how well-supported are the cleaned reference answers themselves? This is an upper bound on what any extractive generator could achieve under our retrieval. The reference set scores 0.748 in-scope faithfulness, 0.826 answer relevance, 0.810 context relevance, 44.2% hallucination, and 10/10 correctly refused negatives, consistent with the Haiku-judged 0.739 and confirming the two judges agree to within 0.01. Walking the gain through the retrieval stages: the original baseline (small corpus, noisy refs, dense  $k=8$ ) sat at Haiku-judged faithfulness of 0.34. The full crawl plus reference audit at dense  $k=4$  already moves it to 0.575. Hybrid alone does not help (0.565). Adding the cross-encoder reranker takes it to 0.739 Haiku / 0.748 Sonnet, with per-answer hallucination dropping from 80.9% to 44.2%.

Second, we ran an actual generator comparison on the cluster (`onj_fri` partition) with the same retrieval, judge, and cleaned set (Table 2). `Llama-3.1-8B-Instruct` comes out on top at 0.665 in-scope faithfulness, 50.0% hallucination, 7/10 correctly refused negatives. `Mistral-7B-Instruct-v0.3` is essentially tied on faithfulness (0.653) and slightly better on context relevance but slower. `GaMS3-12B-Instruct`, despite being trained for Slovenian, came out lowest in faithfulness (0.592) with the highest hallucination rate (66.7%) and only 6/10 correct refusals. The gap between the best generator (0.665) and the reference-grounding ceiling (0.748) is small: under our retrieval and judge, most of the remaining headroom is on the retrieval side rather than the generator side.

### Human evaluation

As a complementary check we conducted a human 0–10 scoring of system outputs on the full 61-question set (51 in-scope, 10 negative), with a human rater comparing each answer against the reference. All four RAG configurations use `Llama-3.1-8B-Instruct` as the generator with hybrid retrieval. The three variables are (i) the embedding model

(`multilingual-e5-base` vs. `BAAI/bge-m3`), (ii) the number of chunks passed to the LLM ( $k \in \{4, 6\}$ ), and (iii) the cross-encoder reranker candidate pool size ( $k_r \in \{20, 30, 50\}$ ). Table 3 shows the results. Scores were revised after auditing the reference answer set: Q10 and Q41 had inverted correct answers, Q43 had transposed phone numbers, and Q13, Q30, Q37 and Q56 lacked references entirely. Claude achieves 8.6/10 mean (revised from 9.1). The best RAG configuration is `bge-m3` with  $k=6$ ,  $k_r=30$  at 5.1/10, followed by `bge-m3` with  $k=4$ ,  $k_r=20$  at 4.8/10. Given this result, we revise our system selection to `bge-m3` at  $k=6$ ,  $k_r=30$  as the final operating setting. Comparing the two default-hyperparameter configurations ( $k=4$ ,  $k_r=20$ ), `bge-m3` still outperforms `e5-base` (4.8 vs. 4.2), matching the automated sweep ordering. A fifth configuration, `bge-m3` with  $k=6$ ,  $k_r=50$ , was also evaluated. Per-question numerical scores were not retained for that run, but a qualitative comparison (keyword coverage, weak-retrieval rate, and per-question improvements and regressions relative to the `e5-base`  $k_r=50$  baseline) is available in the supplementary file `rag_evaluation_report_bge.md`. Note that this human evaluation was conducted on 60 questions (Q44 removed as a duplicate) and uses a slightly different reference set from the LLM-as-judge sweep. The gap between the best RAG system (5.1) and Claude (8.6) remains substantial, consistent with the automated judge ceiling (0.665 vs. 0.748).

### Extended manual evaluation on the full development question set

To stress-test the final operating configuration more broadly, we ran a second manual evaluation on the full 147-question development set (122 in-scope, 25 out-of-scope probes) using `Llama-3.1-8B-Instruct` with hybrid retrieval,  $k=5$  retrieved chunks, and a reranker pool of  $k_r=30$ , at temperature  $T=0$ . This is the same retrieval mode and generator as the best human-evaluation setup, but with a slightly smaller  $k$  (5 vs. 6) and a question set roughly  $2.5\times$  larger. Each answer was graded manually against the reference answer and the retrieved chunks; the 25 out-of-scope probes (weather, gas price, cryptocurrency, recipes, citizenship, CV writing, programming-language advice, restaurant recommendations, etc.) were judged on whether the model correctly refused or incorrectly provided an answer.

Table 4 shows the results. For in-scope questions the system is correct on 97/122 (79.5%), a higher hit-rate than the 5.1/10 mean in the smaller human evaluation. The difference reflects two factors. The human evaluation used a stricter 0–10 partial-credit rubric that captures quality gradations within correct answers, while this pass uses binary grading. The larger question set also includes many more straightforward factual look-ups (contact emails, deadlines, program structures) where the model tends to do well. The out-of-scope refusal rate of 40% (10/25) is lower than the 7/10 (70%) reported by the automated Sonnet judge on the 10-negative subset, mainly because this extended probe set is more diverse

**Table 1.** Embedder × retrieval-mode sweep on `questions_full.v2.json` (Haiku judge, top- $k=4$ ,  $T=0$ ). **Faithfulness:** fraction of answer claims supported by the retrieved chunks ( $[0, 1]$ , higher is better). **Hallucination:** per-answer rate of at least one unsupported claim (lower is better). **Ctx rel.:** judge-rated relevance of retrieved chunks to the question ( $[0, 1]$ ). **Overall:** integer rubric score in  $[1, 5]$ .

| Embedder                    | Mode                   | Faithfulness | Hallucination | Ctx rel.     | Overall (1–5) |
|-----------------------------|------------------------|--------------|---------------|--------------|---------------|
| multilingual-e5-base        | dense                  | 0.575        | 59.1%         | 0.725        | 3.000         |
| multilingual-e5-base        | hybrid                 | 0.565        | 63.6%         | 0.715        | 2.841         |
| <b>multilingual-e5-base</b> | <b>hybrid + rerank</b> | <b>0.739</b> | <b>54.5%</b>  | <b>0.857</b> | <b>3.432</b>  |
| multilingual-e5-large       | dense                  | 0.680        | 54.5%         | 0.812        | 3.250         |
| multilingual-e5-large       | hybrid                 | 0.650        | 61.4%         | 0.787        | 3.205         |
| multilingual-e5-large       | hybrid + rerank        | 0.708        | 59.1%         | 0.866        | 3.364         |

**Table 2.** Generator comparison on the cluster (Sonnet judge,  $T=0$ , same retrieval, top- $k=4$ ). Same metric definitions as Table 1.  $n_{is}$  is the in-scope sample after the judge separates refusals (lower means the generator refused more often). **Refused / 10** counts correctly refused out-of-corpus negatives. The bottom row is the reference-grounding ceiling: Sonnet judging the cleaned references themselves against retrieved chunks;  $n=54$  here because 7 in-scope v2 questions added after the audit have no written reference answer.

| Generator                          | $n_{is}$ | Faithfulness | Hallucination | Ctx rel.     | Overall      | Refused / 10 |
|------------------------------------|----------|--------------|---------------|--------------|--------------|--------------|
| GaMS3-12B-Instruct                 | 54       | 0.592        | 66.7%         | 0.688        | 2.907        | 6/10         |
| <b>Llama-3.1-8B-Instruct</b>       | 52       | <b>0.665</b> | <b>50.0%</b>  | 0.707        | <b>3.000</b> | <b>7/10</b>  |
| Mistral-7B-Instruct-v0.3           | 51       | 0.653        | 58.8%         | <b>0.729</b> | 2.961        | 7/10         |
| <i>Reference-grounding ceiling</i> | 43       | 0.748        | 44.2%         | 0.810        | 3.605        | 10/10        |

and harder: it includes queries where retrieved chunks contain tangentially related content (investment funds, building floor plans with “reception”, film screenings at AGRFT) that the model uses to construct a plausible-sounding but off-topic answer rather than refusing.

The 19 in-scope errors fall into four recurring patterns. **Wrong-domain retrieval** (9 cases): the reranker ranks a doctoral-programme regulation above an undergraduate one (Q2, Q27), surfaces a 2010 annual report with the old faculty address *Tržaška cesta 25* instead of the current *Večna pot 113* (Q104, Q126), retrieves FKKT laboratory-hours data instead of FRI library opening hours (Q8), or pulls a master’s programme document when the question concerns a bachelor’s programme (Q80, Q84). **Misidentified question intent** (4 cases): Q36 asks how long completed lab exercises *remain valid* but the model reports the number of contact *hours*; Q114 confuses “first-year students cannot be tutors” with “there is no tutoring *for* first-year students”; Q62 confuses the home-institution Transcript of Records (printed by the student from STUDIS) with the host-institution document; Q23 confuses a confirmation-of-application email with the official enrollment certificate. **Logical self-contradiction** (1 case): Q61 (4th-exam payment) begins with “No payment needed” but immediately cites the regulation stating all students must pay for 4th and 5th attempts. **Incomplete answer from poor retrieval** (5 cases): Q9 returns only two student statuses drawn from an Erasmus eligibility clause; Q14 names only the two interdisciplinary doctoral programmes FRI co-runs with other faculties, missing the

primary *Računalništvo in informatika* doctorate; Q69 reports conflicting percentage thresholds from two different documents without resolving the number of credit points required for 3rd-year enrollment.

On the out-of-scope side, the 12 failures confirm the pattern noted in the smaller evaluation: the model answers rather than refuses whenever retrieved chunks contain any lexically related text. A query about *baking bread at home* retrieves a food-science laboratory manual and produces oven-temperature advice. A query about *Slovenian citizenship* retrieves an enrollment guide for EU citizens and constructs an answer about applying to the university. A query about *the best programming language in 2025* retrieves course descriptions that mention Python and recommends it. Three out-of-scope questions receive ambiguous answers: for *the best laptop under 1000 EUR* the model correctly states FRI does not publish specific hardware recommendations, for *restaurant near FRI* it recommends the on-campus FRI/FKKT cafeteria (which is in the corpus and factually correct), and for *driving licence procedure* it produces a confused non-answer stating that a driving licence is not required for university admission. Overall, this extended evaluation confirms the automated judge findings and makes the two main failure modes more concrete: wrong-domain retrieval and out-of-scope compliance each need a dedicated mitigation (programme-scope metadata filtering and a query-level refusal guard) before the system is production-ready.

**Table 3.** Human 0–10 evaluation on 60 questions (Q44 removed as duplicate; scorable count varies by configuration as some questions lacked a reference answer). **Claude (direct)** calls Sonnet with no retrieval, as a knowledge ceiling. All RAG variants use Llama-3.1-8B-Instruct and hybrid retrieval.  $k$  = chunks to the LLM;  $k_r$  = reranker candidate pool. \* = default hyperparameters ( $k=4$ ,  $k_r=20$ ) from the automated sweep (Table 1). **Hall.** = answers scoring 0/10. Scores revised after reference-answer audit.

| System                                   | Emb.    | $k$      | $k_r$     | Mean       | $\geq 7$   | $\leq 2$  | Hall.    |
|------------------------------------------|---------|----------|-----------|------------|------------|-----------|----------|
| Claude direct (no RAG)                   | –       | –        | –         | <b>8.6</b> | <b>80%</b> | <b>3%</b> | <b>1</b> |
| <b>RAG (bge-m3, <math>k_r=30</math>)</b> | bge-m3  | 6        | 30        | <b>5.1</b> | 41%        | 28%       | 10       |
| RAG (bge-m3)*                            | bge-m3  | <b>4</b> | <b>20</b> | 4.8        | 40%        | 32%       | 11       |
| RAG (e5-base)                            | e5-base | 6        | 50        | 4.7        | 28%        | 23%       | 6        |
| RAG (e5-base)*                           | e5-base | 4        | 20        | 4.2        | 34%        | 40%       | 12       |

**Table 4.** Extended manual evaluation results on the full 147-question development set (Llama-3.1-8B-Instruct, hybrid + rerank,  $k=5$ ,  $k_r=30$ ,  $T=0$ ). In-scope: 122 questions. Out-of-scope (OOS): 25 questions. **Correct refusal:** model explicitly declines to answer. **Partial:** answer is directionally right but incomplete or vague. **Incorrect:** factually wrong or misidentified answer.

| Category              | Correct           | Partial           | Incorrect         |
|-----------------------|-------------------|-------------------|-------------------|
| In-scope ( $n=122$ )  | 97 (79.5%)        | 6 (4.9%)          | 19 (15.6%)        |
| OOS probes ( $n=25$ ) | 10 refusals (40%) | 3 ambiguous (12%) | 12 answered (48%) |

## Discussion

Both the retrieval-only and the LLM-as-judge results indicate the same pattern. The largest gains were attributable to two stages: corpus expansion and the Sonnet audit moved dense- $k=4$  faithfulness from 0.34 to 0.575 (about 58% of the total gain), and the cross-encoder reranker increased it further to 0.748 (the remaining 42%). Hybrid retrieval on its own produced negligible improvement; it became beneficial only after reranking was applied, presumably because the union of BM25 and dense candidates provided a more diverse candidate pool for the cross-encoder. Another unexpected finding was that multilingual-e5-base outperformed multilingual-e5-large once reranking was applied while embedding approximately  $\sim 2.5\times$  faster, suggesting the cross-encoder absorbed work that would otherwise require a stronger first-stage embedder.

The generator results add another dimension. The best generator (Llama-3.1-8B-Instruct, 0.665 faithfulness) leaves only a 0.083 gap to the reference-grounding ceiling (0.748), meaning that under our retrieval and judge, more than half of the remaining headroom is on the retrieval side rather than the generator side. GaMS3-12B-Instruct being the weakest also surprised us. It is the largest model and the only one trained specifically for Slovenian, but it hallucinated more often (66.7% per-answer rate) and refused fewer negatives (6/10) than the smaller multilingual baselines. Looking at individual questions explains the pattern: on the negative “*Recept za potico z orehi*”, GaMS produces a recipe rather than refusing, while Llama and Mistral correctly say the corpus has no such information. On the in-scope “*Kdaj poteka vpis v višji letnik*”, Llama answers verbatim from a single chunk (faithfulness 1.00) while GaMS adds details not present in any retrieved chunk (faithfulness 0.55, hallucination

flagged). GaMS seems to prefer fluent Slovenian narration over strict context fidelity, which the judge penalises. We did not investigate whether this is prompt-format sensitivity, instruction-tuning data, or something else.

The human evaluation surfaced some clear per-question differences. On Q43 (student-office contact details), the bge-m3 variant ( $k_r=30$ ) is the only one to retrieve the chunk listing all three programme-level e-mail addresses and all three phone numbers correctly, scoring 9/10 (revised upward after the phone-number reference was corrected), while the e5-base ( $k_r=50$ ) run returns only one address (3/10). On Q45 (location of reception and photocopier), the results partly reverse: e5-base ( $k=4$ ,  $k_r=20$ ) scores 9/10 by retrieving the correct building-plan chunk, bge-m3 ( $k_r=30$ ) scores 0/10 by surfacing a document about renting office space, bge-m3 ( $k_r=20$ ) scores 3/10 with a wrong building address (Tržaška cesta 25), and e5-base ( $k_r=50$ ) scores 1/10 because the model places the reception on both the ground floor and the third floor in the same answer. On Q27 (conditions for repeating the first year), all four RAG systems score 0/10 because every run retrieves a doctoral-programme regulation instead of the undergraduate one. The cross-encoder assigns a high relevance score to doctoral credit thresholds (10 KT, 20 KT) even though the question concerns a bachelor-level rule. Claude, which encountered the undergraduate policy text during pre-training, answers 9/10. A different kind of failure shows up on Q61 (“Prevedi ta stavek v japonščino,” a request to translate into Japanese): the model produces an actual Japanese translation instead of refusing, scoring 0/10. This is arguably more harmful than a vague “I don’t know,” because it gives the user a plausible-looking answer to a query the system was never designed to handle. The underlying pattern is the same across all these examples: when the reranker sur-

faces the wrong chunk the generator amplifies the error, and when no relevant chunk exists the model sometimes answers anyway rather than refusing.

Direct comparison against prior university-RAG work [5, 6, 7] is difficult because their corpora and metrics differ (English admissions FAQs, accuracy on closed-set MCQs, etc.), so absolute numbers are not comparable. For this reason we limit ourselves to within-system comparisons (retrieval ablation, embedder sweep, generator sweep) and report all configurations under one consistent judge. The main caveat is the evaluation set size: 51 in-scope and 10 negative questions give limited statistical power, the language is mostly Slovenian, and even at  $T=0$  a single LLM judge has per-question variability. A stratified manual review with Cohen’s  $\kappa$  against the Sonnet judge would be a natural next step.

### Possible improvements

Several improvements follow directly from the failure analysis. **Larger evaluation set.** The current 61-question set gives low statistical power. Running the best configuration on 150–200 questions would make the mean scores and hallucination rates substantially more reliable without changing the pipeline. **Document-level metadata filtering.** The Q27 failure (doctoral regulations retrieved for a bachelor-level query) points to a straightforward fix: tag each chunk with its programme scope (*undergraduate*, *doctoral*, *general*) at index time and hard-filter candidates before the reranker. This would eliminate the entire class of wrong-domain retrievals at no query-time cost. **Out-of-scope detection.** The Q61 failure (model translates into Japanese instead of refusing) calls for a lightweight query classifier (a small intent classifier or a prompt-level guard) that intercepts requests unrelated to university information before they reach the retrieval stage. **Surface-form disambiguation.** The Q45 failure (“reception” matched to “rental premises”) suggests adding document-type or section-header metadata to chunk representations so the cross-encoder can distinguish procedural documents from building-plan pages, reducing false positives from lexical overlap. **LLM-based image parsing.** Our OCR pass recovers text from scanned pages but discards most figures and diagrams. We manually processed the building floor-plan file with an LLM and stored the result as a structured Markdown document for indexing, which is why `e5-base` ( $k=4$ ,  $k_r=20$ ) was able to answer Q45 correctly. Applying the same step systematically to all image-bearing pages in the corpus would extend this coverage to timetable graphics, organisational charts, and similar non-text content.

### Code and data availability

The complete pipeline, cleaned evaluation set, FAISS index, all judge/driver scripts, and SLURM launchers are released at the project repository (branch `eval-pipeline`). Seed URLs are in `raw_dataset/data_links.txt`. The `raw/` and `parsed.jsonl` directories are regenerable. We train no models. All embedders, the reranker, and the generators are open-weight models on Hugging Face Hub, and the

judge uses Anthropic’s API. Only `ANTHROPIC_API_KEY` and `HF_TOKEN` are needed. `code/QUICKSTART.md` is the reproduction recipe.

## References

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